



Predicting commodity returns with climate variables: Statistical loss functions vs. economic value[☆]

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ABSTRACT

We test whether large-scale climate indicators contribute to the predictability of commodity futures returns and generate economic value for investors. Using monthly data on fourteen commodities, we conduct a pseudo out-of-sample forecasting exercise based on a range of models that include automatic variable selection and nonlinear regime-switching specifications. Climate-related predictors rarely outperform standard benchmarks in terms of statistical forecast accuracy. Yet, when embedded in a portfolio choice problem, they deliver economically meaningful gains. These results illustrate a disconnect between statistical predictability and economic relevance.

1. Introduction

Commodity futures represent a major asset class, offering diversification benefits, inflation hedging, and equity-like long-run returns (Gorton and Rouwenhorst, 2006). Despite their relevance, evidence on short-horizon return predictability in commodity futures remains mixed, and the nature of the underlying priced risks is debated. It is unclear whether commodity futures returns are primarily driven by a common stochastic discount factor or by specific sources of risk (De Roon et al., 2000).

Climate shocks are a natural candidate for such commodity risks because they impact directly production, inventories, and transportation constraints, thereby shifting marginal costs and convenience yields (Yang et al., 2026). A prominent example is represented by *teleconnections*, i.e., persistent atmospheric oscillations linking geographically distant regions. Heino et al. (2018) document that teleconnections affect global cropland areas and discuss their relevance for commodity markets. Among large-scale climate phenomena, the El Niño–Southern Oscillation represents a salient source of persistent and geographically heterogeneous climate shocks, with documented effects on both spot and futures prices, particularly for agricultural commodities. More broadly, climate-driven shocks propagate beyond commodity markets, affecting global inflation dynamics and economic activity (Cashin et al.,

2017). Recent work in finance further shows that physical climate risk can be priced in asset markets, although most contributions focus on equities rather than commodities (Choi et al., 2020; Kruttli et al., 2025).

Nonetheless, it is unclear whether large-scale climate indicators provide information beyond macroeconomic factors that can be exploited to improve short-horizon forecasts of commodity futures returns and, crucially, whether they generate economically meaningful gains when embedded in portfolio problems and, more generally, in actual financial decisions (Tedeschi et al., 2024). Our paper addresses this gap by jointly evaluating whether climate information adds to macroeconomic predictors in terms of predictive performance and/or whether, even when statistical predictability is weak, this translates into risk-adjusted gains to investors.

We use monthly data on futures returns for a cross-section of fourteen commodities over the period January 1989 to May 2018, collected from Thomson Reuters Eikon. Climate information is summarized using PCs extracted from a broad set of large-scale atmospheric and climate indicators, which are included alongside macroeconomic factors in the information set.

Our design relies on three classes of forecasting models. First, we estimate predictive regressions in which predictors are chosen recursively from a large set using stepwise selection guided by information criteria,

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allowing the specification to adapt to evolving predictor relevance and parameter instability. Second, we extend these baseline models with regime-dependent features indexed to market volatility, so that both the set of included predictors and their predictive content may differ across states (DeMiguel et al., 2009; Cederburg et al., 2023). Third, we employ nonlinear models based on hidden Markov dynamics, which allow regression coefficients to vary across latent regimes and capture state dependence in the relationship between climate conditions and commodity returns (Ang and Timmermann, 2012).

Forecasts are generated at a one-month horizon within a recursive pseudo-out-of-sample design and evaluated relative to an AR(1) benchmark. To assess economic relevance, the same forecasts are embedded in a portfolio allocation exercise for a power-utility investor, following DeMiguel et al. (2009). Performance is measured using certainty-equivalent returns, i.e., the constant return that delivers the same average utility as the realized of portfolio returns. Our results convey a clear message. When forecast accuracy is assessed solely by statistical loss, climate-based predictors rarely outperform the benchmarks. However, when forecasts are mapped into portfolio decisions, climate signals generate meaningful gains, even when their improvements under conventional loss functions are modest. This evidence highlights a wedge between statistical predictability and economic value in commodity markets when climate information is used, consistent with the literature on forecast evaluation and economic performance (Cenesizoglu and Timmermann (2012)).

2. Research design

We study the predictability of commodity futures returns using a recursive framework. For each commodity j , we consider the one-month-ahead predictive regression

$$R_{t+1}^j = \alpha_t^j + \mathbf{Z}_t' \beta_t^j + \varepsilon_{t+1}^j, \quad (1)$$

where R_{t+1}^j denotes excess return between t and $t+1$ and $\mathbf{Z}_t$ is a vector of lagged predictors observed at time t . Model parameters are estimated using an expanding window. Following, e.g., Dal Pra et al. (2018), forecasts are generated using a recursive, pseudo out-of-sample design. At each forecast origin, the estimation sample is expanded by one observation and the model is re-estimated using only information available up to time t . The pseudo out-of-sample evaluation period spans from December 2003 to April 2018.

Forecast accuracy is evaluated relative to a benchmark AR(1) model,

$$R_{t+1}^j = \alpha_t^j + \theta_t^j R_t^j + \nu_{t+1}^j, \quad (2)$$

which captures the (weak) autocorrelation structure typically observed in commodity futures returns. Statistical forecasting performance is assessed using the root mean squared forecast error (RMSFE).¹

2.1. Principal component analysis

The predictor set includes macroeconomic variables and climate-based indicators. Given the potentially large dimensionality of such a set, we adopt a sparse forecasting approach in which relevant predictors are selected recursively using stepwise regression procedures based on the Akaike Information Criterion, implemented through both forward and backward selection.

Climate information is summarized using principal component analysis applied to a vector $\mathbf{C}_t \in \mathbb{R}^n$ collecting n large-scale atmospheric and climate indicators. The k th climate principal component is defined as

$$PC_{k,t} = \mathbf{u}_k' \mathbf{C}_t, \quad k = 1, \dots, K. \quad (3)$$

¹ We also test forecast accuracy using mean absolute error. The results are qualitatively similar to those reported and are available upon request.

We retain the first $K = 5$ principal components (PCs), which jointly explain over 50% of the total variance of the climate information set. The selected climate PCs enter the regressions as candidate predictors and are subject to the same recursive variable selection procedure as the other regressors. For consistency, macroeconomic information is summarized using the first eight principal extracted from a large panel of U.S. macroeconomic and financial variables.

2.2. Nonlinear models and regime dependence

To allow for potential state dependence in return predictability, we consider two complementary classes of models. First, stepwise predictive regressions are estimated allowing for regime dependence based on market volatility, distinguishing between high- and low-volatility periods.² Second, we estimate nonlinear predictive models based on hidden Markov dynamics (Ang and Timmermann, 2012), in which coefficients vary across latent regimes:

$$R_{t+1}^j = \alpha_{s_t}^j + \mathbf{Z}_t' \beta_{s_t}^j + \varepsilon_{t+1}^j, \quad s_t \in \{1, \dots, S\}, \quad (4)$$

where the latent state s_t follows a first-order Markov process. Markov switching models capture abrupt changes in the conditional mean dynamics and provide a flexible representation of nonlinear predictability.

2.3. Economic value and portfolio allocation

To assess the economic relevance of the forecasts, we embed the predictive signals into a recursive portfolio allocation problem for a myopic investor with power utility in which, at each time t , the investor chooses portfolio weights ω_t to solve

$$\max_{\omega_t} \mathbb{E}_t \left[\frac{(1 + \omega_t' \mathbf{R}_{t+1})^{1-\gamma}}{1-\gamma} \right] \quad \text{s.t.} \quad \omega_t' \mathbf{1} = 1, \quad \omega_{i,t} \geq 0, \quad (5)$$

where γ is the coefficient of relative risk aversion, $\mathbf{R}_{t+1}$ collects the returns on commodities, equities, bonds, and cash. The climate-based predictive signals are employed exclusively for commodities, while forecasts for equities and bonds are obtained using the benchmark models.

Following DeMiguel et al. (2009), portfolio performance is evaluated using the certainty equivalent return (CER). Let $r_{p,t}$ denote the realized return of strategy p and $r_{b,t}$ the return of the AR(1) benchmark. The out-of-sample certainty equivalent return ΔCER_p is defined as:

$$\frac{1}{T} \sum_{t=1}^T (1 + r_{p,t})^{1-\gamma} = \frac{1}{T} \sum_{t=1}^T [(1 + \Delta CER_p)(1 + r_{b,t})]^{1-\gamma}. \quad (6)$$

We also consider an alternative specification incorporating transaction costs proportional to portfolio rebalancing.

2.4. Data

We consider monthly excess returns on fully collateralized futures contracts for a cross-section of 14 commodities over the period January 1989 to May 2018.³ Data are obtained from Thomson Reuters Eikon. The commodity universe includes agricultural commodities (corn, soybeans, wheat, cocoa, coffee, cotton, and sugar), soft commodities (frozen orange juice, lumber, and live cattle), energy (light crude oil),

² Market volatility regimes are identified using the VIX index as a proxy for aggregate uncertainty. Specifically, we estimate a two-state Markov-switching model for the VIX and classify each period as belonging to a high- or low-volatility regime based on the filtered regime probabilities.

³ Details on data sources and summary statistics are provided in Online Appendix A.

and precious metals (gold, silver, and platinum). For each commodity j , returns are computed at a monthly frequency as

$$R_{t+1}^j = \frac{F_{t+1}^j}{F_t^j} - 1, \quad (7)$$

where F_t^j denotes the end-of-month futures price. Positions are systematically rolled into the second-nearest contract to avoid delivery effects.

Climate information is based on a set of 13 large-scale atmospheric and climate indicators capturing persistent global climate variability. The set includes major teleconnection indices describing large-scale atmospheric circulation patterns (such as ENSO-related measures and Pacific and Atlantic oscillations), together with global indicators of drought conditions and temperature anomalies. These variables are common across commodities and observed at a monthly frequency. Climate data are obtained from standard sources, including the National Oceanic and Atmospheric Administration and its Climate Prediction Center, as well as the National Centers for Environmental Information. All series are transformed to ensure stationarity prior to analysis. All series are transformed to ensure stationarity prior to analysis. Appendix A provides a detailed description of the climate variables and their data sources.

Macroeconomic information is summarized using PCs extracted from a large monthly dataset compiled by Ludvigson and Ng (2009), which includes 132 U.S. macroeconomic and financial time series spanning real activity, labor markets, prices, interest rates, and financial market indicators.

3. Results

3.1. Baseline out-of-sample forecasting performance

As regards the stepwise linear predictive regressions, out-of-sample RMSFE are reported in Table 6 of Online Appendix B. While stepwise variable selection reduces forecast errors relative to the kitchen-sink specification, which suffers from severe overfitting, none of the models outperform the AR(1) benchmark. Moreover, the inclusion of climate factors does not yield systematic gains in forecast accuracy.

To investigate heterogeneity across commodities, Table 1 classifies assets into climate-exposed (CE) and less climate-exposed (LCE) groups. Climate-exposed commodities include agricultural, soft, and livestock commodities, as well as lumber, whose returns are more directly affected by weather and climate conditions. Less climate-exposed commodities include energy and precious metals, whose exposure to climate variability is less direct. This classification allows us to assess whether climate variables provide stronger predictive gains for commodities that are more directly exposed to weather conditions.

Table 2 reports out-of-sample RMSFE by periods classified by volatility regimes, using the filtered probabilities of a regime switching model applied to the VIX. In low-volatility periods, stepwise regressions outperform the AR(1) benchmark for most commodities (12 out of 14), with gains largely driven by macroeconomic specifications (10 out of 12 cases). Climate variables provide additional improvements over macro-only models in only three cases. In high-volatility periods, forecast improvements largely vanish: stepwise regressions rarely outperform AR(1), and climate-augmented models do not deliver systematic gains.

Table 3 reports RMSFE for the HMM specifications. Except for live cattle, where the macro-based HMM marginally improves upon the benchmark, HMMs fail to outperform the AR(1) model in nearly all cases. Importantly, as before, augmenting the HMM with climate variables does not lead to lower RMSFE relative to the macro-only specification.

Climate-exposed commodities do not exhibit systematically better forecast accuracy than less exposed assets; rather, their forecast errors

Table 1

Classification of commodities by climate exposure.

Climate-Exposed Commodities (CE)	Less Climate-Exposed Commodities (LCE)
Corn	Light (Crude Oil)
Soybeans	Gold
Wheat	Silver
Coffee	Platinum
Cocoa	
Sugar	
Cotton	
FOJ	
Lumber	
Live Cattle	

Commodities are classified based on their exposure to climate-related shocks, following Heino et al. (2018) and Yang et al. (2026). Climate-exposed commodities include agricultural, soft, and livestock commodities, as well as lumber, whose production is directly affected by weather and climate conditions. Less climate-exposed commodities include energy and precious metals, which are either indirectly affected or largely insensitive to climate variability in the short run.

display greater dispersion across models and specifications. Nonetheless, a directionally consistent pattern emerges in the HMM results (Table 3): augmenting the macro-only HMM with climate factors reduces average RMSFE for climate-exposed commodities (from 0.0942 to 0.0933) while increasing it for less climate-exposed ones (from 0.0949 to 0.1014), suggesting that climate information is at least not detrimental — and marginally useful — for the group whose returns are more directly tied to weather conditions.

3.2. Economic value

Table 4 reports annualized ΔCER relative to the AR(1) benchmark.⁴ In the absence of transaction costs, the Hidden Markov Model augmented with macroeconomic and climate factors delivers the highest economic value across all levels of risk aversion, with ΔCER equal to 0.56%, 0.68%, and 0.86% for $\gamma = 2, 5$, and 10, respectively. By contrast, the macro-only HMM generates smaller gains, while stepwise strategies yield negative ΔCER throughout.

These gains arise despite the lack of systematic improvements in forecast accuracy: as shown earlier, climate-based models rarely outperform the AR(1) benchmark in terms of RMSFE. This evidence is in line with the literature documenting a disconnect between predictive performance and economic value, whereby models with weak or modest forecast accuracy can generate economically meaningful gains when evaluated in a portfolio context (Dal Pra et al., 2018). Although stepwise strategies remain unprofitable, the inclusion of climate information consistently mitigates their losses relative to macro-only specifications.

Panel B shows that these conclusions are robust to the inclusion of transaction costs. Overall, climate risk appears to affect optimal portfolio allocation and risk exposures to deliver higher realized risk-adjusted performance rather than short-horizon return statistical predictability.⁵ We report bootstrap confidence intervals for CER differences in Table 7 (Appendix B). While these intervals are generally wide and often include zero, in several cases the confidence intervals for climate-augmented HMM specifications do not overlap with those of the corresponding benchmarks.⁶

⁴ Additional portfolio diagnostics, including mean returns, volatility, Sharpe ratios, and downside for the benchmark and climate-augmented strategies, are reported in Online Appendix B (Table 6).

⁵ These conclusions are robust to the inclusion of commodity-specific (e.g., basis, momentum, illiquidity, hedging intensity) and aggregate commodity factors. Detailed results are available upon request.

⁶ In addition, the HMM with macro and climate factors yields p-values that are in some cases close to conventional significance levels.

Table 2

RMSFE — Stepwise regressions by volatility regimes.

	Light	Corn	Soyb.	Wheat	Coffee	Cocoa	Sugar	Cotton	Gold	Silver	Plat.	FOJ	Lumber	Cattle	Avg CE	Avg LCE
<i>Low volatility</i>																
Kitchen Sink	0.1012	0.0750	0.0766	0.0925	0.1177	0.0841	0.0834	0.0766	0.0519	0.1719	0.0631	0.1118	0.0964	0.0537	0.0868	0.0970
Stepwise Macro (Backward)	0.0767	0.0729	0.0751	0.0818	0.0899	0.0706	0.0709	0.0709	0.0452	0.0818	0.0539	0.0854	0.0756	0.0497	0.0743	0.0644
Stepwise Macro (Forward)	0.0762	0.0730	0.0751	0.0818	0.0914	0.0706	0.0709	0.0688	0.0453	0.0803	0.0538	0.0756	0.0756	0.0493	0.0732	0.0639
Stepwise Macro + Climate (Backward)	0.0779	0.0731	0.0760	0.0814	0.0896	0.0708	0.0723	0.0717	0.0460	0.0835	0.0546	0.0862	0.0756	0.0500	0.0747	0.0655
Stepwise Macro + Climate (Forward)	0.0787	0.0724	0.0751	0.0820	0.0915	0.0708	0.0723	0.0694	0.0462	0.0821	0.0545	0.0853	0.0756	0.0497	0.0744	0.0654
<i>High volatility</i>																
Kitchen Sink	0.1737	0.1624	0.1909	0.2907	0.2175	0.2531	0.2350	0.1502	0.1301	0.1853	0.2252	0.1403	0.1409	0.0840	0.1865	0.1786
Stepwise Macro (Backward)	0.1667	0.1600	0.1468	0.1347	0.0971	<u>0.1434</u>	0.1187	0.1080	0.0868	0.1506	<u>0.1129</u>	0.1125	<u>0.0960</u>	0.0436	0.1161	0.1293
Stepwise Macro (Forward)	0.1172	0.1555	0.1468	<u>0.1323</u>	0.0956	0.1444	0.1127	0.1029	0.0687	0.1242	0.1147	0.1088	0.0961	<u>0.0424</u>	0.1138	0.1062
Stepwise Macro + Climate (Backward)	0.1682	0.1546	0.1484	0.1749	<u>0.0950</u>	0.1458	0.1169	0.1081	0.0871	0.1534	0.1150	0.1146	0.0986	0.0465	0.1203	0.1309
Stepwise Macro + Climate (Forward)	0.1180	<u>0.1523</u>	0.1485	0.1395	<u>0.0956</u>	0.1469	0.1148	0.1033	0.0691	0.1248	0.1159	0.1156	0.1004	0.0429	0.1150	0.1070
AR(1)	<u>0.0844</u>	<u>0.0869</u>	<u>0.0798</u>	<u>0.0948</u>	0.0877	<u>0.0784</u>	<u>0.0844</u>	<u>0.0809</u>	<u>0.0521</u>	<u>0.0961</u>	<u>0.0728</u>	<u>0.0915</u>	<u>0.0804</u>	0.0465	0.0811	0.0764

The table reports RMSFE obtained for low- and high-volatility regimes. Boldfaced (underlined) entries indicate the lowest RMSFE within the low- (high-) volatility regime across stepwise specifications. The low-volatility periods correspond to January 2004–December 2007 and January 2013–May 2018, while the high periods span January 2008–December 2012. The columns “Avg CE” and “Avg LCE” report the average RMSFE for climate-exposed (Corn, Soybeans, Wheat, Coffee, Cocoa, Sugar, Cotton, Frozen Orange Juice, Lumber, and Live Cattle) and less climate-exposed commodities (Light Crude Oil, Gold, Silver, and Platinum), respectively.

Table 3

Root mean square forecast error — Hidden Markov models.

	Light	Corn	Soyb.	Wheat	Coffee	Cocoa	Sugar	Cotton	Gold	Silver	Plat.	FOJ	Lumber	Cattle	Avg CE	Avg LCE
HMM — Macro Factors	0.0911	0.0934	0.0870	0.1038	0.1441	0.0934	0.0959	0.0972	0.0960	0.1078	0.0848	0.0937	0.0867	0.0465	0.0942	0.0949
HMM — Macro and Climate Factors	0.1448	0.0928	0.0976	0.1035	0.1092	0.0965	0.0907	0.1022	0.0627	0.1104	0.0875	0.1015	0.0910	0.0480	0.0933	0.1014
AR(1)	0.0844	0.0869	0.0798	0.0948	0.0877	0.0784	0.0844	0.0809	0.0521	0.0961	0.0728	0.0915	0.0804	0.0465	0.0811	0.0764

The table reports RMSFE from a recursive pseudo out-of-sample exercise for HMM. Boldfaced entries indicate the lowest RMSFE for each commodity. The columns “Avg CE” and “Avg LCE” report the average RMSFE for climate-exposed (Corn, Soybeans, Wheat, Coffee, Cocoa, Sugar, Cotton, Frozen Orange Juice, Lumber, and Live Cattle) and less climate-exposed commodities (Light Crude Oil, Gold, Silver, and Platinum), respectively.

The emerging divergence between statistical and economic performance reflects a fundamental difference between statistical loss functions and economic evaluation criteria. Standard measures such as RMSFE assess forecast accuracy in terms of the conditional mean and penalize forecast errors symmetrically, without accounting for their implications for decision-making. By contrast, portfolio choice depends on how predictive signals affect time-varying portfolio weights and the associated risk exposures, and therefore on the economic relevance—rather than the statistical magnitude—of forecast errors. Even modest improvements in predictive signals may have a limited impact on average forecast accuracy, while still affecting the investor's optimal allocation through changes in expected returns, higher-order moments, or the covariance structure of returns. As a result, relatively small or noisy signals can generate economically meaningful gains if they help investors adjust portfolio tilts or manage risk exposures more effectively over time.

A formal quantitative bridge between these two perspectives is provided by [Campbell and Thompson \(2008\)](#), who show that, for a mean–variance investor, even an out-of-sample R^2 as small as 0.5% at a monthly horizon is sufficient to generate economically meaningful improvements in portfolio performance. The intuition is that the achievable gain in certainty-equivalent return scales with the signal-to-noise ratio of the predictive model rather than with its contribution to average squared forecast error. Our results are broadly consistent with this logic: the climate-augmented HMM delivers annualized ΔCER in the range of 56–86 basis points (Panel A of [Table 4](#)) despite near-zero or negative out-of-sample R^2 values across most commodities.⁷

⁷ The magnitude of these gains appears to exceed what a linear R^2 -to-Sharpe mapping would imply under the [Campbell and Thompson \(2008\)](#) framework, pointing to the regime-switching structure as an additional source of economic value: climate signals that carry limited information about the unconditional mean forecast error can nonetheless shift optimal portfolio weights across latent states, reducing exposure in adverse regimes and raising it when expected returns are more favorable.

Table 4

Economic value.

	$\gamma = 2$	$\gamma = 5$	$\gamma = 10$
<i>Panel A: No transaction costs</i>			
HMM — Macro Factors	0.0023	0.0026	0.0041
HMM — Macro and Climate Factors	0.0056	0.0068	0.0086
Stepwise Macro (Forward)	−0.0071	−0.0099	−0.0190
Stepwise Macro and Climate (Forward)	−0.0067	−0.0096	−0.0184
AR(1) Benchmark	0.0000	0.0000	0.0000
<i>Panel B: With transaction costs</i>			
HMM — Macro Factors	0.0025	0.0052	0.0053
HMM — Macro and Climate Factors	0.0079	0.0045	0.0128
Stepwise Macro (Forward)	−0.0031	−0.0052	−0.0153
Stepwise Macro and Climate (Forward)	−0.0009	−0.0014	−0.0090
AR(1) Benchmark	0.0000	0.0000	0.0000

The table reports annualized CERs relative to an AR(1) benchmark for different levels of relative risk aversion. Panel A reports results without transaction costs, while Panel B accounts for proportional costs. Boldfaced entries indicate the highest ΔCER for each level of risk aversion.

4. Robustness

We test the robustness of results to alternative constructions of climate-based predictors, with detailed results in online Appendix C. While recursive and orthogonalized climate factors do not improve forecast accuracy, the HMM with orthogonalized macro-climate factors delivers higher certainty equivalent returns than the macro-only model.

5. Conclusion

We study the role of climate in forecasting commodity futures returns and in generating economic value. Climate predictors fail to materially improve short-horizon prediction accuracy relative to benchmarks. However, when incorporated into portfolio choice, climate information delivers economically significant gains, consistent with prior evidence that economic value need not coincide with improvements in statistical accuracy. These findings suggest that climate risk affects optimal portfolios rather than short-horizon predictability, highlighting

Table 5
Climate variables: Definition and data sources.

Variable	Climate phenomenon	Sample	Data source
Pacific North American (PNA)	Atmospheric circulation over the North Pacific and North America	1989–2018	NOAA Climate Prediction Center
Arctic Oscillation (AO)	Strength of the polar vortex (Northern Hemisphere)	1989–2018	NOAA Climate Prediction Center
North Atlantic Oscillation (NAO)	Sea-level pressure gradient over the North Atlantic	1989–2018	NOAA Climate Prediction Center
Scandinavian Pattern (SCA)	Geopotential height anomalies over Scandinavia and Europe	1989–2018	NOAA Climate Prediction Center
Pacific Decadal Oscillation (PDO)	Low-frequency North Pacific sea surface temperature variability	1989–2018	NOAA National Centers for Environmental Information
Atlantic Multidecadal Oscillation (AMO)	Multidecadal North Atlantic sea surface temperature anomalies	1989–2018	NOAA Physical Sciences Laboratory
Multivariate ENSO Index (MEI)	Coupled ocean–atmosphere ENSO indicator	1989–2018	NOAA Physical Sciences Laboratory
Dipole Mode Index (DMI)	Indian Ocean Dipole sea surface temperature gradient	1989–2018	NOAA Physical Sciences Laboratory
Madden–Julian Oscillation (MJO, phase 1)	Intra-seasonal tropical convection pattern	1989–2018	World Meteorological Organization
Quasi-Biennial Oscillation (QBO proxy)	Stratospheric circulation variability (temperature-based proxy)	1989–2018	NOAA Physical Sciences Laboratory
Palmer Drought Severity Index (global, 36 m MA)	Global drought conditions (soil moisture balance)	1989–2018	National Center for Atmospheric Research
Northern Hemisphere Snow Cover Anomaly	Cryosphere variability (snow extent deviations)	1989–2018	NOAA National Centers for Environmental Information
Global Temperature Anomaly (detrended)	Global surface temperature anomalies (land and ocean)	1989–2018	NOAA Physical Sciences Laboratory

All variables are observed at monthly frequency and standardized prior to analysis. The Palmer Drought Severity Index is averaged globally and smoothed using a 36-month moving average. Global temperature anomalies are spatially averaged and detrended to remove the deterministic warming component.

Table 6
Portfolio diagnostics.

	$\gamma = 2$				$\gamma = 5$				$\gamma = 10$			
	Mean	Vol	Sharpe	Downside	Mean	Vol	Sharpe	Downside	Mean	Vol	Sharpe	Downside
<i>Panel A: No transaction costs</i>												
HMM — Macro Factors	0.0159	0.0851	0.1755	0.0590	0.0159	0.0851	0.1755	0.0590	0.0159	0.0851	0.1759	0.0590
HMM — Macro and Climate Factors	0.0183	0.0775	0.2239	0.0518	0.0183	0.0775	0.2237	0.0518	0.0183	0.0775	0.2236	0.0518
Stepwise Macro (Forward)	0.0080	0.0915	0.0766	0.0680	0.0080	0.0915	0.0766	0.0680	0.0080	0.0915	0.0765	0.0680
Stepwise Macro and Climate (Forward)	0.0085	0.0915	0.0824	0.0678	0.0085	0.0915	0.0824	0.0678	0.0085	0.0915	0.0824	0.0678
AR(1) Benchmark	0.0137	0.0844	0.1509	0.0595	0.0137	0.0844	0.1509	0.0595	0.0137	0.0844	0.1509	0.0595
<i>Panel B: With transaction costs</i>												
HMM — Macro Factors	0.0140	0.0809	0.1617	0.0567	0.0161	0.0802	0.1882	0.0550	0.0135	0.0772	0.1629	0.0540
HMM — Macro and Climate Factors	0.0181	0.0711	0.2413	0.0470	0.0150	0.0758	0.1849	0.0520	0.0168	0.0688	0.2303	0.0455
Stepwise Macro (Forward)	0.0099	0.0882	0.1013	0.0645	0.0110	0.0885	0.1137	0.0642	0.0081	0.0807	0.0878	0.0595
Stepwise Macro and Climate (Forward)	0.0119	0.0867	0.1265	0.0620	0.0147	0.0880	0.1557	0.0619	0.0133	0.0808	0.1529	0.0568
AR(1) Benchmark	0.0115	0.0794	0.1323	0.0567	0.0115	0.0798	0.1317	0.0570	0.0112	0.0784	0.1306	0.0562

The table reports standard portfolio diagnostics for the main strategies, including mean returns, volatility, Sharpe ratios, and downside risk. Panel A reports results without transaction costs, while Panel B accounts for costs.

Table 7
Statistical inference on economic value: Parametric bootstrap.

	$\gamma = 2$		$\gamma = 5$		$\gamma = 10$	
	Δ CER	95% CI	Δ CER	95% CI	Δ CER	95% CI
<i>Panel A: No transaction costs</i>						
HMM — Macro	+0.0023	[−0.1234, +0.1712]	+0.0026	[−0.1374, +0.1814]	+0.0041	[−0.1859, +0.2337]
HMM — Macro & Climate	+0.0056	[−0.0747, +0.2119]	+0.0068	[−0.0662, +0.2463]	+0.0086	[−0.0861, +0.3314]
Stepwise Macro (Fwd)	−0.0071	[−0.2454, +0.0671]	−0.0099	[−0.2864, +0.0622]	−0.0190	[−0.4427, +0.0957]
Stepwise Macro & Climate (Fwd)	−0.0067	[−0.2380, +0.0774]	−0.0096	[−0.2875, +0.0744]	−0.0184	[−0.3923, +0.0953]
<i>Panel B: With transaction costs</i>						
HMM — Macro	+0.0025	[−0.1089, +0.1658]	+0.0052	[−0.0999, +0.2004]	+0.0053	[−0.1955, +0.2357]
HMM — Macro & Climate	+0.0079	[−0.0373, +0.2265]	+0.0045	[−0.0878, +0.2084]	+0.0128	[−0.0318, +0.3343]
Stepwise Macro (Fwd)	−0.0031	[−0.1903, +0.1062]	−0.0052	[−0.2210, +0.1091]	−0.0153	[−0.3034, +0.1448]
Stepwise Macro & Climate (Fwd)	−0.0009	[−0.1577, +0.1387]	−0.0014	[−0.1909, +0.1686]	−0.0090	[−0.1968, +0.1969]

The table reports annualized Δ CER relative to the AR(1) benchmark and 95% bootstrap confidence intervals based on a parametric bootstrap with $B = 10,000$ simulations. The DGP is AR(0)–GJR–GARCH(1,1), calibrated separately for each specification.

the importance of complementing statistical forecast evaluation with economic loss function-driven analyses.

Appendix A. Climate variables

See Table 5.

Appendix B. Portfolio diagnostics

See Table 6.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econlet.2026.113028>.

Data availability

Data will be made available on request.

References

- Ang, Andrew, Timmermann, Allan, 2012. Regime changes and financial markets. *Annu. Rev. Financ. Econ.* 4 (1), 313–337.
- Campbell, John Y., Thompson, Samuel B., 2008. Predicting excess stock returns out of sample: Can anything beat the historical average? *Rev. Financ. Stud.* 21 (4), 1509–1531.
- Cashin, Paul, Mohaddes, Kamiar, Raissi, Mehdi, 2017. Fair weather or foul? The macroeconomic effects of El Niño. *J. Int. Econ.* 106, 37–54.
- Cederburg, Scott, Johnson, Travis L., O'Doherty, Michael S., 2023. On the economic significance of stock return predictability. *Rev. Financ.* 27 (2), 619–657.
- Cenesizoglu, Tolga, Timmermann, Allan, 2012. Do return prediction models add economic value? *J. Bank. Financ.* 36 (11), 2974–2987.
- Choi, Darwin, Gao, Zhenyu, Jiang, Wenxi, 2020. Attention to global warming. *Rev. Financ. Stud.* 33 (3), 1112–1145.
- Dal Pra, Giulia, Guidolin, Massimo, Pedio, Manuela, Vasile, Fabiola, 2018. Regime shifts in excess stock return predictability: an out-of-sample portfolio analysis. *J. Portf. Manag.* 44 (3), 10.
- De Roon, Frans A., Nijman, Theo E., Veld, Chris, 2000. Hedging pressure effects in futures markets. *J. Financ.* 55 (3), 1437–1456.
- DeMiguel, Victor, Garlappi, Lorenzo, Uppal, Raman, 2009. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Rev. Financ. Stud.* 22 (5), 1915–1953.
- Gorton, Gary, Rouwenhorst, K. Geert, 2006. Facts and fantasies about commodity futures. *Financ. Anal. J.* 62 (2), 47–68.
- Heino, Matias, Puma, Michael J, Ward, Philip J, Gerten, Dieter, Heck, Vera, Siebert, Stefan, Kumm, Matti, 2018. Two-thirds of global cropland area impacted by climate oscillations. *Nat. Commun.* 9 (1), 1257.
- Krutli, Mathias S., Roth Tran, Brigitte, Watugala, Sumudu W., 2025. Pricing poseidon: extreme weather uncertainty and firm return dynamics. *J. Financ.* 80 (2), 783–832.
- Ludvigson, Sydney C., Ng, Serena, 2009. Macro factors in bond risk premia. *Rev. Financ. Stud.* 22 (12), 5027–5067.
- Tedeschi, Marco, Foglia, Matteo, Bouri, Elie, Dai, Peng-Fei, 2024. How does climate policy uncertainty affect financial markets? Evidence from europe. *Econom. Lett.* 234, 111443.
- Yang, Hao, Yang, Jie, Feng, Yun, 2026. Climate physical risks and the vulnerability of global agricultural commodities. *Econom. Lett.* 258, 112748.